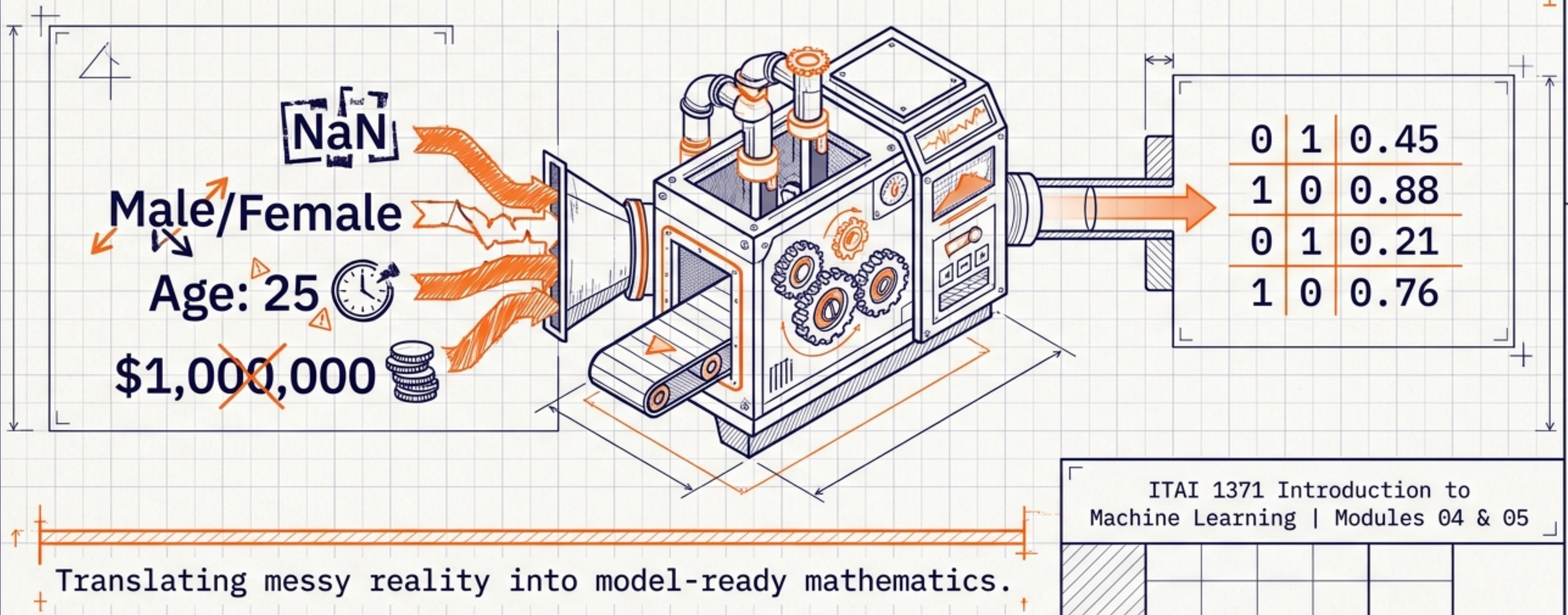


The Data Refinery Blueprint

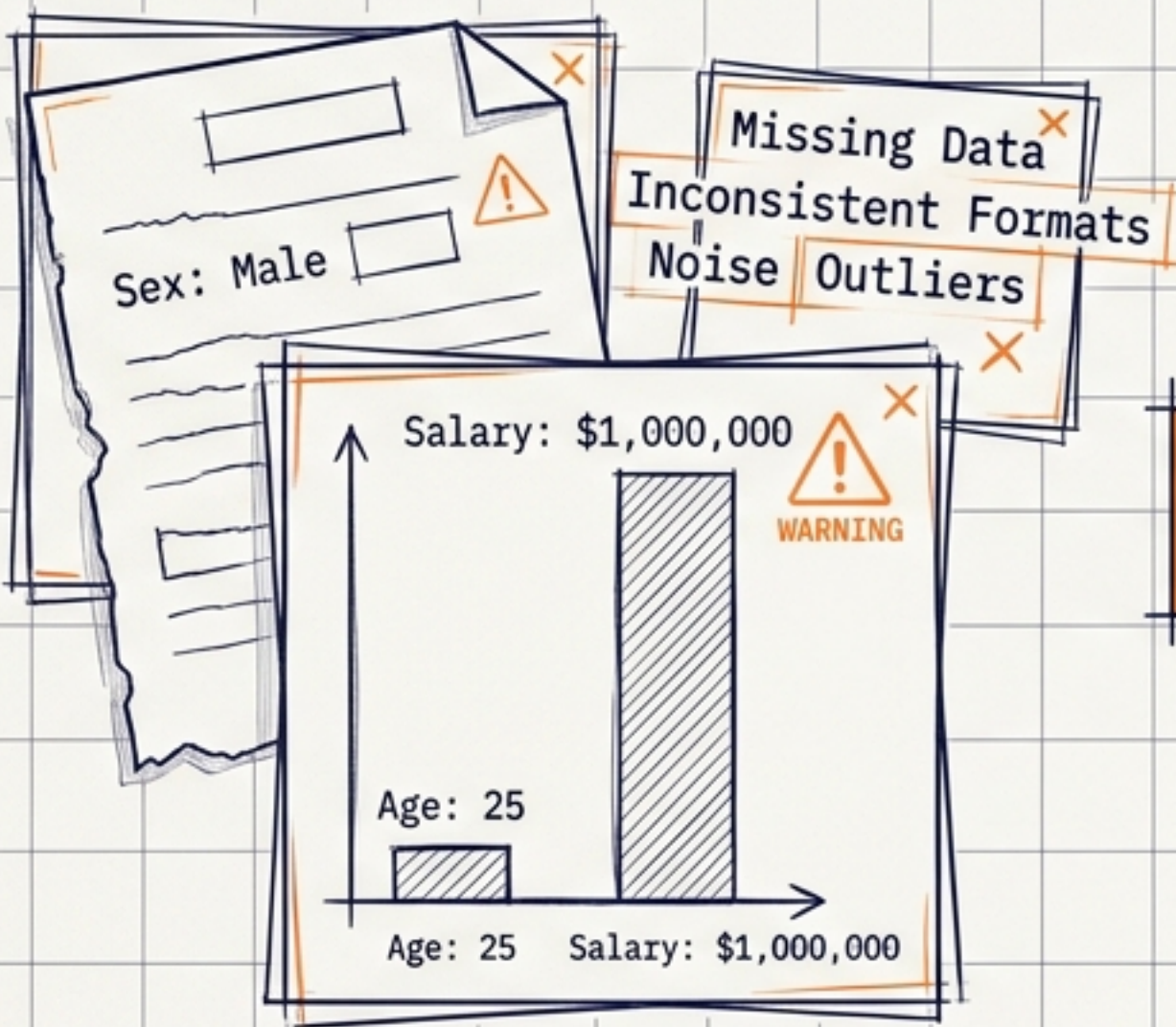
End-to-End Exploratory Data Analysis & Preparation



ITAI 1371 Introduction to
Machine Learning | Modules 04 & 05

Models Only Speak Mathematics

The Reality Check



The Translation

0	1	0.25	1000000.0
1	0	0.30	50000.0
0	1	0.22	75000.0
1	0	0.45	200000.0
1	1	0.25	20000.0
0	1	0.25	30000.0
1	0	0.22	10000.0
1	1	0.30	50000.0
1	0	0.25	25000.0
0	1	0.25	100000.0
1	0	0.30	50000.0
0	1	0.22	75000.0
1	0	0.25	50000.0
1	0	0.45	200000.0

The Translation Problem:

ML algorithms rely on addition, multiplication, and dot products. They cannot interpret text strings.

Garbage In, Garbage Out:

A simple model on clean data beats a complex model on messy data.

The 80/20 Rule:

80% of real-world machine learning work happens in the data preparation phase.

The 6-Stage EDA Assembly Line



PROJECT: EDA PIPELINE SCHEMATIC		
DRANN BY: ENG. DEPT.		
DATE: OCT 26, 2024		
SHEET: 2 OF 6		

1. Setup & Ingestion

2. Summary Statistics

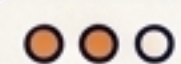
3. Missing Data RCA

4. Univariate Analysis

5. Bivariate/Multivariate

6. Insights & Action Plan

Stations 1 & 2: Ingestion and Summary

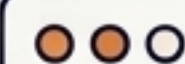


Structure (Setup)

`df.head()` | `df.info()` | `df.shape`

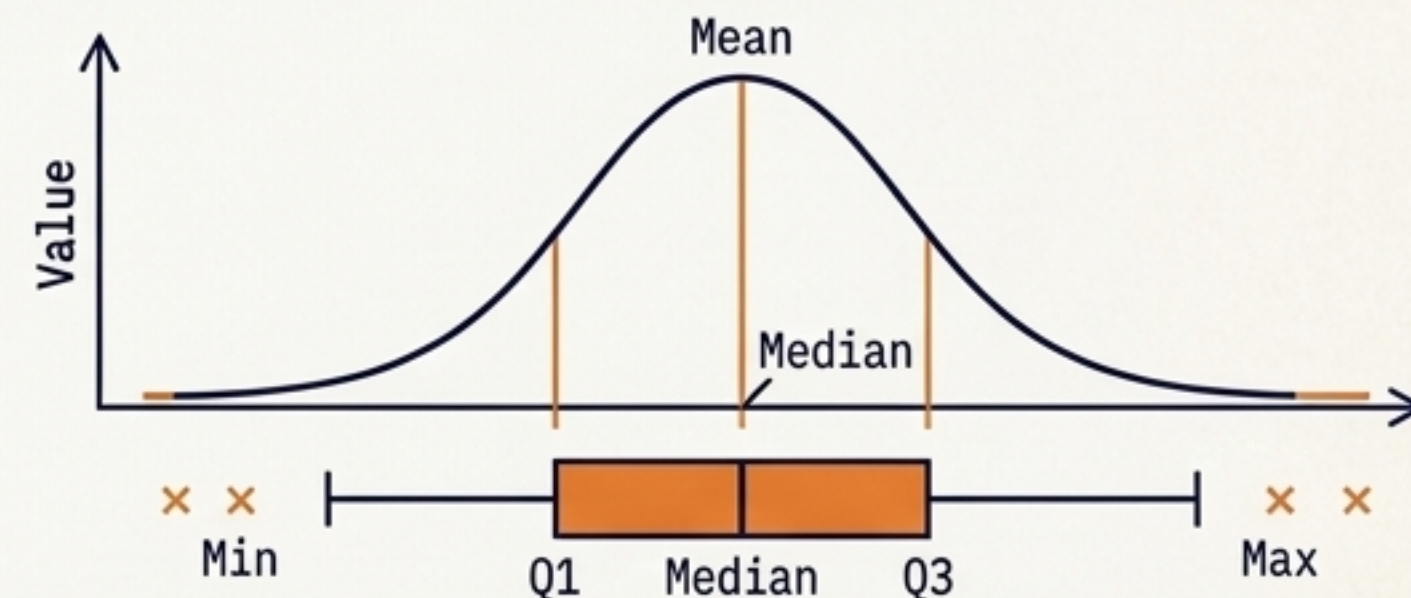
	ID (int)	Name (object)	Age (int)	Salary (float)	Date (object)
	1	Rran Name 1	25	1,000,000	2024-02-11
	2	Rran Name 2	25	1,000,000	2024-12-22
	3	Rrae Same 3	25	1,000,000	2024-12-23
	...	...	...	...	...
	...	...	...	...	...

Verifies basic health. What are the column names? What is the data type? How many rows exist?



Central Tendency (Summary)

`df.describe()` | `df['col'].value_counts()`



Evaluates mathematical spread (mean, median, standard deviation). Immediately flags impossible minimums (e.g., Age = -5).

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1. Setup & Ingestion

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Station 3: Missing Data Root Cause Analysis (RCA)

Core Tool: `df.isnull().sum()`

Diagnostic Screen

	ID	Feature A	Feature B	Feature C	Feature D	Feature E
	0					
	1					
	2					
	3					
	4					
	5					

Diagnostic Table

MCAR (Missing Completely at Random)	MAR (Missing at Random)	MNAR (Missing Not at Random)
No pattern. Caused by random system errors. Action: Safe to drop or impute.	Missingness depends on other observed columns (e.g., Age missing only for 3rd class). Action: Impute based on relationships.	Missingness depends on the missing value itself (e.g., high- income earners skip the salary field). Action: Highly dangerous. Do not drop. The absence is a signal.

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1. Setup & Ingestion

2. Summary Statistics

3. Missing Data RCA

4. Univariate Analysis

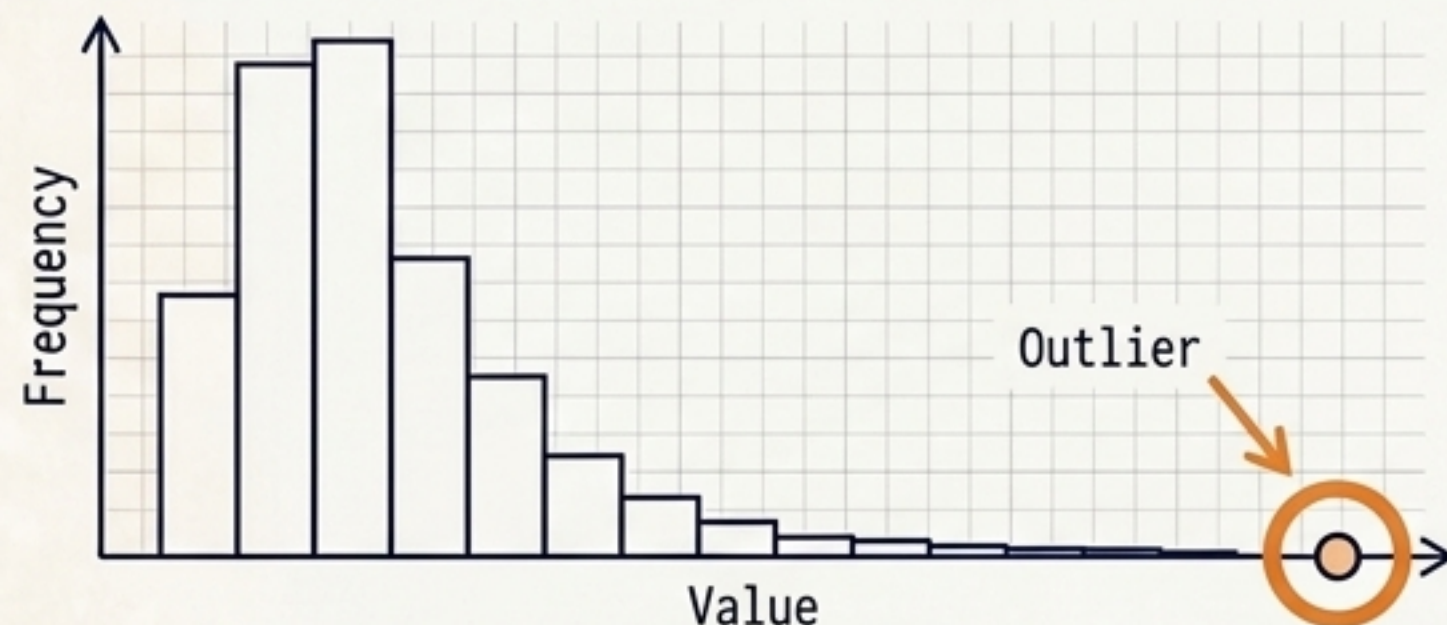
5. Bivariate/Multivariate

6. Insights & Action Plan

Stations 4 & 5: Distributions & Relationships

Univariate Analysis (Isolation)

`sns.histplot()` | `sns.boxplot()`

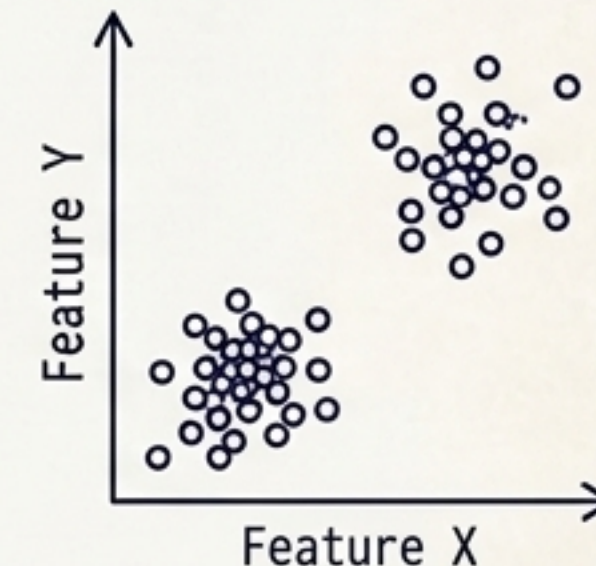


Purpose: Spot skewness (long tails) and dangerous outliers. Determines if data requires transformation to meet model assumptions.

Bivariate & Multivariate (Intersections)

`sns.heatmap(df.corr())` | `sns.scatterplot()`

	Feature A	Feature B	Feature C	Feature D	Target
Feature A	1.0	0.6	0.3	-0.1	0.1
Feature B	-0.85	1.0	0.7	0.3	0.3
Feature C	-0.85	-0.1	1.0	-0.1	-0.1
Feature D	-0.96	-0.32	-0.8	1.0	0.7
Target	-0.85	0.6	-0.1	-0.32	1.0



Purpose: Identify feature-target relationships and feature-feature multicollinearity (redundant data).

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1. Setup & Ingestion

2. Summary Statistics

3. Missing Data RCA

4. Univariate Analysis

5. Bivariate/Multivariate

6. Insights & Action Plan

Station 6: The Titanic Discoveries

Demographics:
Women and Children First



Observation: Female survival rate ~74%. Male survival rate ~18%. Children <10 saw significantly higher survival.

Insight: Sex and Age are the most critical predictive features.

Socioeconomic:
Money Matters



Observation: 1st Class had highest survival; 3rd Class lowest. Fare distribution is highly skewed with a massive long tail.

Insight: Pclass is a primary indicator. Fare requires transformation.

Data Quality:
Gaps in the Record



Observation: Cabin data is ~77% missing. Age is ~20% missing. Embarked is missing 2 values.

Insight: Cabin is too sparse to keep as is.

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The Insight-to-Action Matrix

Observations

Cabin column is 77% missing.

Age column is 20% missing.

Features like Sex and Embarked are text strings.

Fare is massively skewed with high outliers.

Data Preparation Actions

Action: Drop the column entirely, or engineer a binary Has_Cabin indicator flag.

Action: Impute missing values using the Median (robust against outliers) or Advanced KNN.

Action: Apply One-Hot or Label Encoding to translate them into math.

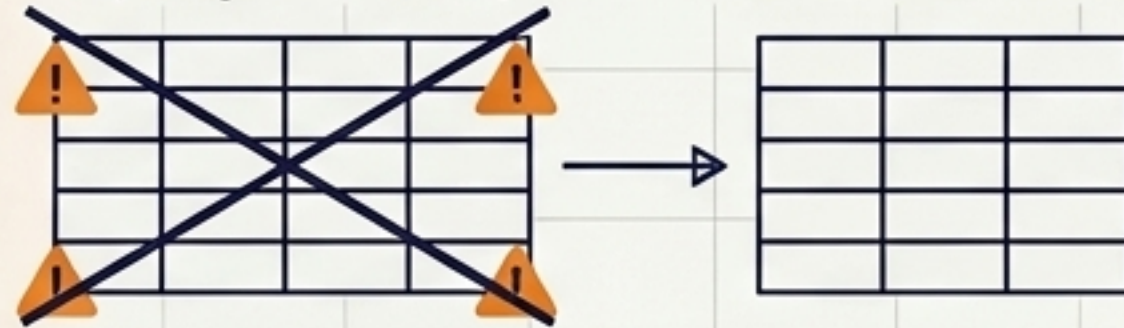
Action: Apply Robust Scaling or Log Transformations.

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Processing Plant: Missing Data Strategies

Deletion (The Nuclear Option)

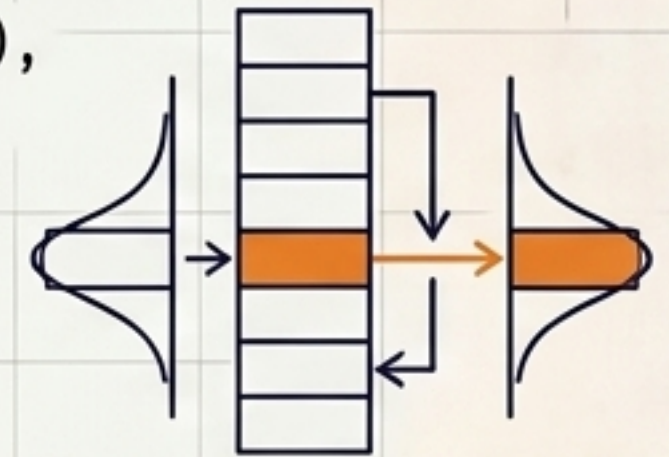
Listwise (Drop Row): Use when <5% missing.
Column (Drop Feature): Use when >70% missing.



Tradeoff: Safe, but destroys valuable information.

Simple Imputation (The Fast Fix)

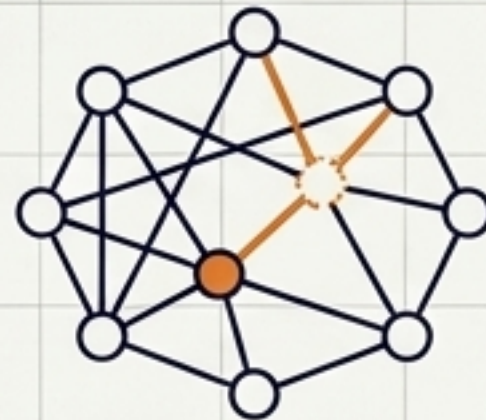
Fill with Mean (normal data),
Median (skewed data), or
Mode (categorical).



Tradeoff: Reduces variance; ignores relationships.

Advanced Imputation (The Algorithmic Fix)

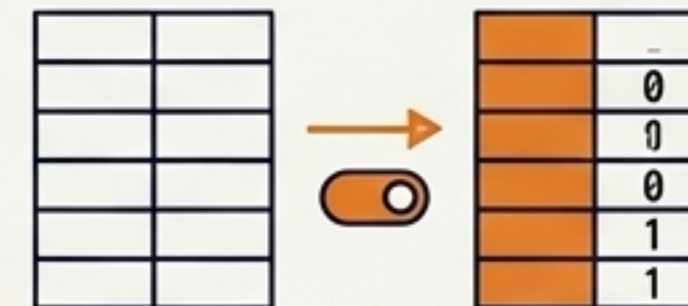
KNN Imputation: Uses
nearest neighbors to
predict the gap.



Tradeoff: Highly accurate, but computationally expensive.

Indicator Features (The Signal Extraction)

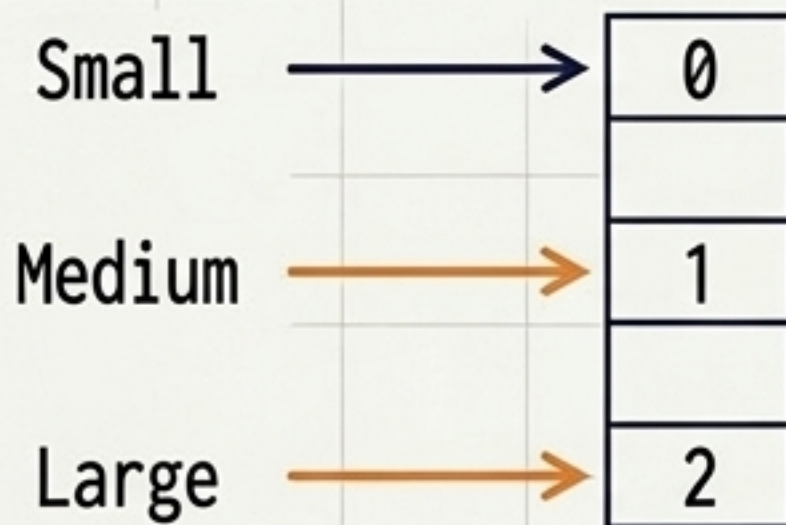
Create a binary column (Age_Missing: 0 or 1).



Tradeoff: Treats the absence of data as predictive information.

Processing Plant: Categorical Translation

Label Encoding (For Ordinal Data)

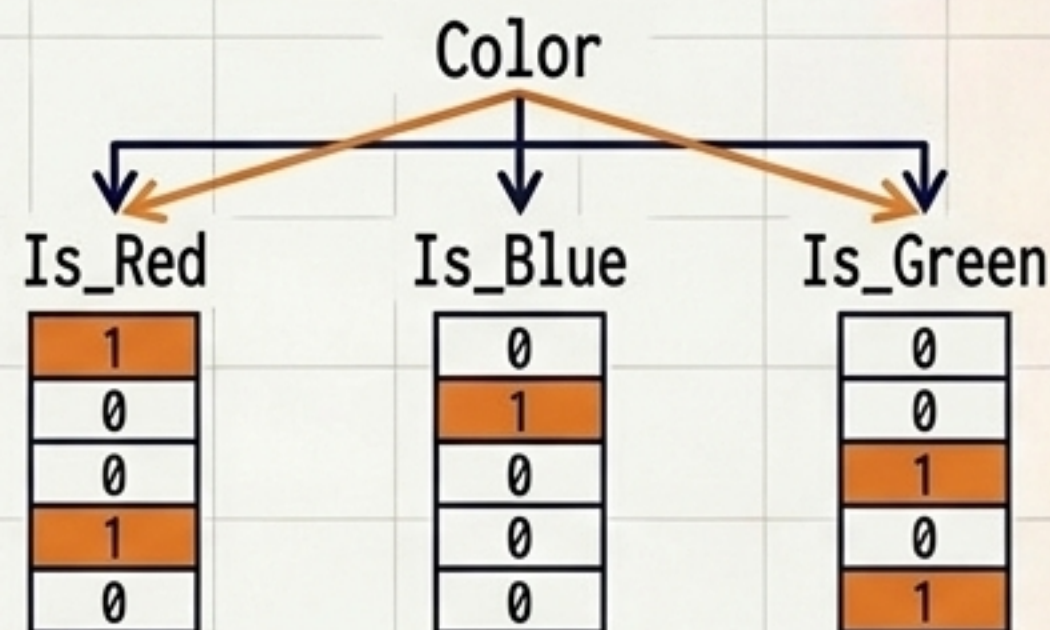


Converts categories to integers.

Rule: Only use when mathematical order matters.

Hazard: Never use for nominal data; the model will falsely assume Paris (2) > London (1).

One-Hot Encoding (For Nominal Data)



Creates a new binary column for every unique category.

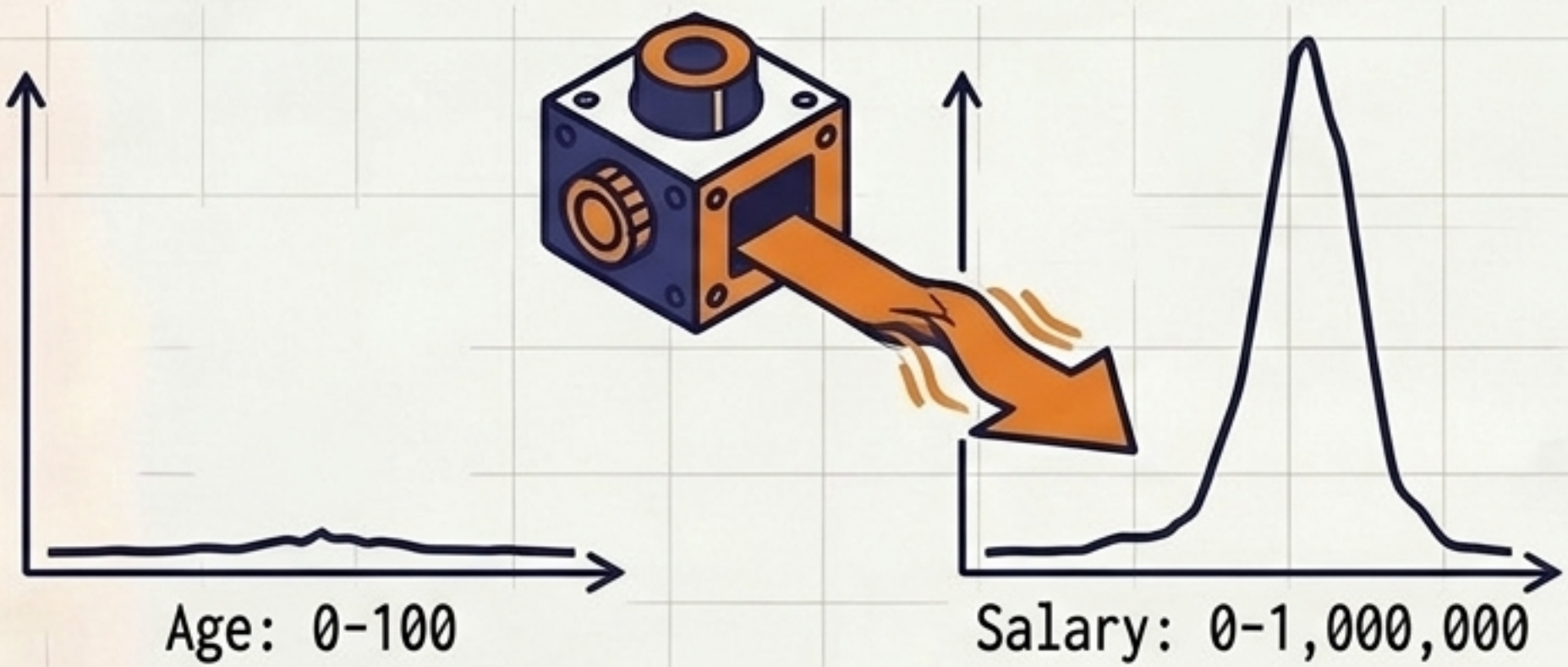
Rule: Use when categories have no inherent rank.

Hazard: High cardinality leads to the Curse of Dimensionality.

! The Dummy Trap Warning: For linear models, always drop one encoded column (drop='first') to prevent multicollinearity. **!**

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Processing Plant: Feature Scaling



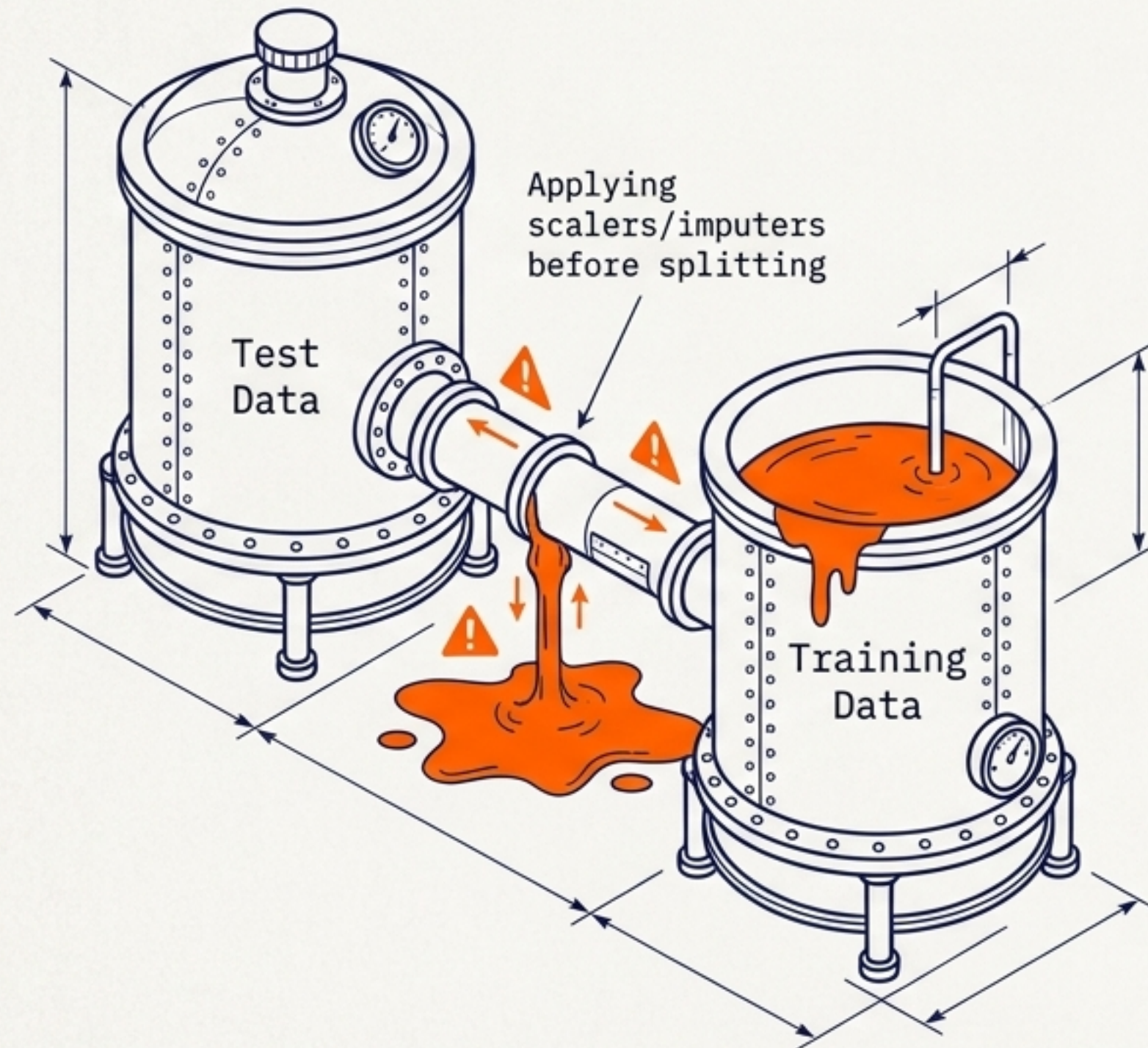
The Distance Bias

Distance-based algorithms (KNN, SVM, K-Means) will be entirely dominated by features with large numbers and ignore features with small numbers.

StandardScaler (Z-Score)	MinMaxScaler (Normalization)	RobustScaler (The Outlier Killer)
Centers data around a mean of 0. Sensitive to massive outliers.	Bounds all data strictly between 0 and 1.	Uses Medians and Interquartile Ranges (IQR) to scale the bulk of the data without letting outliers squash the scale.

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Critical Hazard: Data Leakage



The Silent Killer



Leakage occurs when information from the test set (future data) bleeds into the training process, resulting in mathematically impossible real-world performance (e.g., 100% accuracy).

The Golden Rule Workflow

1. Split your data FIRST.
2. `scaler.fit(X_train)` -> Learn parameters from Training set ONLY.
3. `scaler.transform(X_test)` -> Apply exact rules to Test set. NEVER fit on test data.

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Anatomy of a Model-Ready Dataset

1. 100% Numeric:

Every single string and category has been mathematically encoded.

2. 100% Complete:

Zero missing values (NaNs). All gaps intelligently imputed, dropped, or flagged.

3. Uniformly Scaled:

Magnitudes are balanced so algorithms treat features equitably.

4. Enriched:

New features (like FamilySize or Title) have been engineered from raw variables.

5. Isolated:

Strictly separated into Training, Validation, and Testing sets to prevent leakage.

0.45	-1.2	1	0
0.78	-0.03	0	1
-2.1	0.55	1	0
0.32	-1.1	0	1

PROJECT: EDA PIPELINE SCHEMATIC

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> LAB_05_DATA_PREP.EXE : Deliverables

Mission Objective: Execute the complete Titanic EDA and Preparation pipeline.

Canvas Deliverables (Due Tuesday 15 Sep 2026):

- ☐ 1. Executed Notebook PDF: L04_[Name]_ITAI1371.pdf (All cells run in order, outputs visible).
- ☐ 2. Reflective Journal PDF: 1-2 pages analyzing workflow complexity and learning types.
- ☐ 3. Contribution Journal PDF: 1-2 pages detailing independent efforts.

Execution Note



Implement both Simple and Advanced (KNN) imputation.
Compare Standardization vs. Normalization.
Use the Markdown cells to document the why,
not just the how.

[SYSTEM STATUS: Ready for Processing]

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